

MehtaOmPset4

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Problem Set 4

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FINM 34700

```
[84]: import numpy as np
import pandas as pd
from scipy import stats
from scipy.stats import f
from sklearn.metrics import pairwise_distances
import prince
import matplotlib.pyplot as plt
from sklearn.manifold import MDS
```

1. *Simple MANOVA layout*

a)

$$\bar{x}_1 = \frac{48}{12} = 4$$

$$\bar{x}_2 = \frac{60}{12} = 5$$

$$x_1 = \begin{bmatrix} 6 & 5 & 8 & 4 & 7 \\ 3 & 1 & 2 & & \\ 2 & 5 & 3 & 2 & \end{bmatrix} = 4 \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & & \\ 1 & 1 & 1 & 1 & \end{bmatrix} + \begin{bmatrix} 2 & 2 & 2 & 2 & 2 \\ -2 & -2 & -2 & & \\ -1 & -1 & -1 & -1 & \end{bmatrix} + \begin{bmatrix} 0 & -1 & 2 & -2 & 1 \\ 1 & -1 & 0 & & \\ -1 & 2 & 0 & -1 & \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 7 & 9 & 6 & 9 & 9 \\ 3 & 6 & 3 & & \\ 3 & 1 & 1 & 3 & \end{bmatrix} = 5 \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & & \\ 1 & 1 & 1 & 1 & \end{bmatrix} + \begin{bmatrix} 3 & 3 & 3 & 3 & 3 \\ -1 & -1 & -1 & & \\ -3 & -3 & -3 & -3 & \end{bmatrix} + \begin{bmatrix} -1 & 1 & -2 & 1 & 1 \\ -1 & 2 & -1 & & \\ 1 & -1 & -1 & 1 & \end{bmatrix}$$

b)

```
[95]: x1 = [
        [6, 5, 8, 4, 7],
        [3, 1, 2],
        [2, 5, 3, 2]
    ]

    x2 = [
        [7, 9, 6, 9, 9],
        [3, 6, 3],
        [3, 1, 1, 3]
    ]

    X = []
    groups = []
    for g, (g1, g2) in enumerate(zip(x1, x2), start=1):
        for i in range(len(g1)):
            X.append([g1[i], g2[i]])
            groups.append(g)

    X = np.array(X)
    groups = np.array(groups)

    x_bar = X.mean(axis=0)

    group_means = {}
    for g in np.unique(groups):
        group_means[g] = X[groups == g].mean(axis=0)

    W = np.zeros((2, 2))
    for g in np.unique(groups):
        Xg = X[groups == g]
        mean_g = group_means[g]
        residuals = Xg - mean_g
        W += residuals.T @ residuals

    B = np.zeros((2, 2))
    for g in np.unique(groups):
        n_g = np.sum(groups == g)
        diff = (group_means[g] - x_bar).reshape(2, 1)
        B += n_g * (diff @ diff.T)

    T = W + B
    wilks_lambda = np.linalg.det(W) / np.linalg.det(T)

    np.linalg.det(W), np.linalg.det(B), wilks_lambda
```

[95]: (155.00000000000003, 720.0000000000001, 0.0361895867382676)

d)

```
[96]: p = 2 # number of dependent variables
      g = 3 # number of groups
      n = 12 # total number of observations

      scaling_factor = n - 1 - (p + g) / 2
      chi_square_stat = -np.log(wilks_lambda) * scaling_factor
      df = p * (g - 1)

      # Compute p-value
      p_value = 1 - stats.chi2.cdf(chi_square_stat, df)

      chi_square_stat, df, p_value
```

[96]: (28.21136281591705, 4, 1.1301100103167094e-05)

e)

$$(H_0) : \mu_1 = \mu_2 = \mu_3$$
$$(H_a) : \text{At least one } \mu_t \text{ differs from the others}$$

In words:

The null hypothesis states that the mean vectors of the two responses (i.e., x_1 and x_2) are the same across all three treatment groups. That is, the treatments have no overall multivariate effect on the outcomes. \ The alternative hypothesis states that at least one treatment group differs in its mean vector — meaning there is a significant difference in the combined responses (x_1, x_2) across groups.

Since the p-value obtained from Bartlett's approximation in part (d) is extremely small ($\approx 1.13 \times 10^{-5}$), we reject H_0 . This provides strong evidence that the treatments have significantly different effects on the joint responses.

2. MANOVA and confidence region

```
[97]: df = pd.read_csv('T6-13.DAT', sep=r"\s+", header=None)
      df.columns=['x1', 'x2', 'x3', 'x4', 'period']
      per1, per2, per3 = df.loc[df['period']==1], df.loc[df['period']==2], df.
        ↪loc[df['period']==3]
```

a)

$$p = 4$$

$$g = 3$$

$$n_1 = n_2 = n_3 = 30$$

b)

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

H_a : At least one μ differs from the rest

c)

```
[98]: X = np.array(df)[:,:4]
total_mean = X.mean(axis=0)
labels = np.array(df['period'])
per1 = np.array(per1)[:,:4]
per2 = np.array(per2)[:,:4]
per3 = np.array(per3)[:,:4]
per1_mean = per1.mean(axis=0)
per2_mean = per2.mean(axis=0)
per3_mean = per3.mean(axis=0)
per1_residual = (per1 - per1_mean).T @ (per1 - per1_mean)
per2_residual = (per2 - per2_mean).T @ (per2 - per2_mean)
per3_residual = (per3 - per3_mean).T @ (per3 - per3_mean)
W = per1_residual + per2_residual + per3_residual
n = 30
B1 = n * np.outer(per1_mean - total_mean, per1_mean - total_mean)
B2 = n * np.outer(per2_mean - total_mean, per2_mean - total_mean)
B3 = n * np.outer(per3_mean - total_mean, per3_mean - total_mean)
B = B1 + B2 + B3
wilks = np.linalg.det(W) / np.linalg.det(W + B)
n = 90
p = 4
t1 = (n - p - 2) / p
lambda_root = np.sqrt(wilks)
f_stat = t1 * (1 - lambda_root) / lambda_root
df1 = 2 * p
df2 = 2 * (n - p - 2)
p_value = 1 - f.cdf(f_stat, df2, df1)
wilks, p_value
```

[98]: (0.8301026836053987, 2.1744965631453894e-06)

$$\Lambda^* \approx 0.83$$

$$p \approx 2.17 \cdot 10^{-6}$$

At the 5% significance level, we reject H_0 that all means are equal, as the p-value is effectively 0. This means that at least one time period has significantly different skull measurements.

d)

Assumptions for this F-statistic are **multivariate normality, homogeneity of covariance matrices, independence of observations, independent treatments, and $n \gg p$ for all groups**

e)

$$\binom{g}{2} \cdot p \Rightarrow \binom{3}{2} \cdot 4 = 12$$

This is because you would only be doing pairwise tests **among 3 different groups**, and within the groups there are **4 variables**, each giving a separate confidence interval for the difference in means of each variable in each group.

3. Multidimensional Scaling

```
[99]: with open("T12-13.DAT") as f:
      raw = f.read().split()
      data = np.array([float(x) for x in raw])

      mat = np.zeros((9, 9))
      mat[np.triu_indices(9)] = data
      X = mat.T
      X = X + X.T
```

a)

```
[100]: coords = []
      for q in range(3, 6):
          mds = MDS(n_components=q, dissimilarity='precomputed', random_state=42)
          coords_q = mds.fit_transform(X)
          coords.append(coords_q)
```

```
[101]: for i, q in enumerate(range(3, 6)):
      print(f"\nCoordinates for q = {q} dimensions:")
      df = pd.DataFrame(coords[i], columns=[f"Dim{j+1}" for j in range(q)])
```

```
df.index = [f"Site {k+1}" for k in range(len(coords[i]))]
display(df)
```

Coordinates for q = 3 dimensions:

	Dim1	Dim2	Dim3
Site 1	0.303692	0.041876	0.743028
Site 2	-0.226884	-0.187009	-0.729506
Site 3	-0.050006	0.044434	0.495585
Site 4	-0.252788	0.733109	0.029408
Site 5	-1.196997	-0.073611	-0.172882
Site 6	0.152541	0.180823	0.512839
Site 7	0.147988	0.093412	-0.489832
Site 8	0.593933	-0.330486	-1.205757
Site 9	0.528522	-0.502548	0.817117

Coordinates for q = 4 dimensions:

	Dim1	Dim2	Dim3	Dim4
Site 1	0.236771	0.196308	0.653354	0.343573
Site 2	-0.195262	0.074840	-0.651011	-0.408250
Site 3	0.352628	0.064902	0.306252	0.154284
Site 4	0.039592	-0.649294	0.294356	-0.292802
Site 5	0.798425	-0.301308	-0.816544	-0.223575
Site 6	0.250259	0.262564	0.460954	-0.024642
Site 7	-0.449998	-0.217521	-0.120800	-0.186079
Site 8	-1.144534	0.256611	-0.682633	-0.240390
Site 9	0.112120	0.312898	0.556070	0.877881

Coordinates for q = 5 dimensions:

	Dim1	Dim2	Dim3	Dim4	Dim5
Site 1	-0.373791	0.305722	-0.137550	0.597623	-0.157964
Site 2	0.400671	-0.286404	0.260560	-0.562864	-0.114246
Site 3	-0.264515	0.349733	0.018894	0.228509	-0.049388
Site 4	0.209782	0.352122	-0.328352	-0.050999	0.555357
Site 5	-0.109934	0.475905	0.563750	-0.899244	0.316240
Site 6	-0.112486	0.363578	0.041336	0.342929	-0.227294
Site 7	0.335443	-0.324085	-0.205240	-0.125692	0.142527
Site 8	0.602949	-1.185384	0.006809	-0.309723	-0.186695
Site 9	-0.688120	-0.051186	-0.220207	0.779462	-0.278539

b)

```
[102]: stress_vals = []

for i, q in enumerate(range(3, 6)):
    coords_q = coords[i]                # Embedded coordinates for q
    ↪ dimensions
    d_q = pairwise_distances(coords_q)    #  $d_{ij}^{(q)}$ 

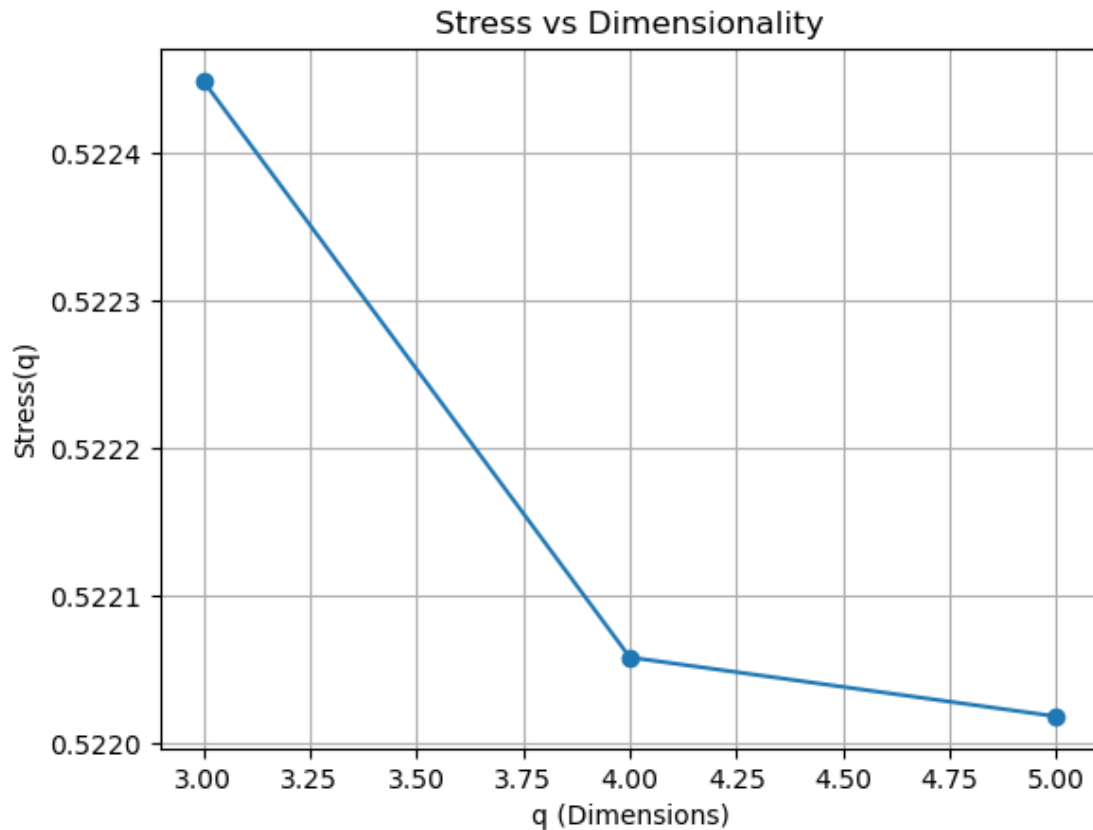
    numerator = np.sum((X - d_q)**2) / 2    # Sum over  $j < i$ 
    denominator = np.sum(X**2) / 2         # Sum over  $j < i$ 
    stress = np.sqrt(numerator / denominator)

    stress_vals.append(stress)
    print(f"Stress({q}) = {stress:.4f}")
plt.plot(range(3, 6), stress_vals, marker='o')
plt.xlabel("q (Dimensions)")
plt.ylabel("Stress(q)")
plt.title("Stress vs Dimensionality")
plt.grid(True)
plt.show()
```

Stress(3) = 0.5224

Stress(4) = 0.5221

Stress(5) = 0.5220



c)

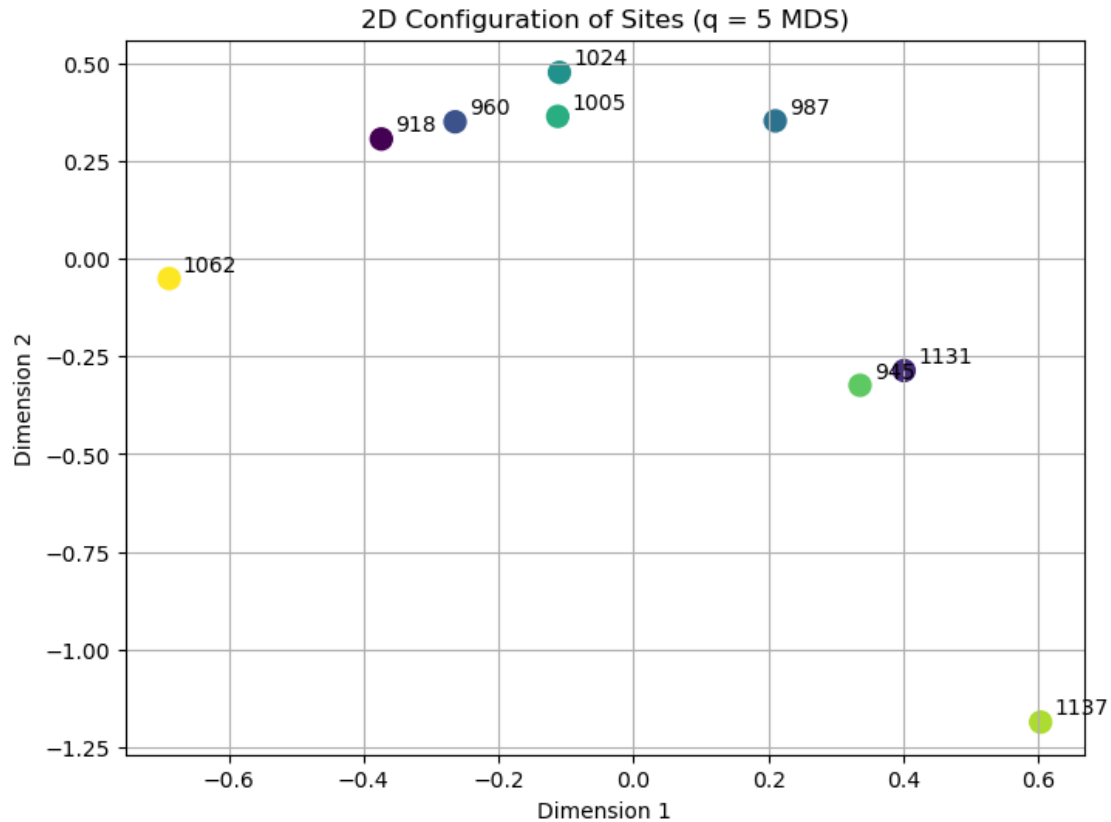
```
[103]: coords_5 = coords[2] # 5D coordinates (q = 5)
periods = [918, 1131, 960, 987, 1024, 1005, 945, 1137, 1062]

x = coords_5[:, 0] # First dimension
y = coords_5[:, 1] # Second dimension

plt.figure(figsize=(8, 6))
plt.scatter(x, y, c=range(9), cmap='viridis', s=100)

for i, period in enumerate(periods):
    plt.text(x[i] + 0.02, y[i] + 0.02, str(period), fontsize=10)

plt.title("2D Configuration of Sites (q = 5 MDS)")
plt.xlabel("Dimension 1")
plt.ylabel("Dimension 2")
plt.grid(True)
plt.show()
```

4. Correspondence Analysis

a)

```
[104]: df = pd.read_csv('HairEyeAll.txt', sep=r'\s+', header=None)
df.index = ['Black', 'Brown', 'Red', 'Blond']
df.columns=['Brown', 'Blue', 'Hazel', 'Green']
df_copy = df.copy()
```

```
[105]: row_pcts = df.div(df.sum(axis=1), axis=0).round(2)
col_pcts = df.div(df.sum(axis=0), axis=1).round(2)
display(row_pcts, col_pcts)
```

	Brown	Blue	Hazel	Green
Black	0.63	0.19	0.14	0.05
Brown	0.42	0.29	0.19	0.10
Red	0.37	0.24	0.20	0.20
Blond	0.06	0.74	0.08	0.13

	Brown	Blue	Hazel	Green
Black	0.31	0.09	0.16	0.08
Brown	0.54	0.39	0.58	0.45

Red	0.12	0.08	0.15	0.22
Blond	0.03	0.44	0.11	0.25

b)

```
[106]: df
```

```
[106]:
```

	Brown	Blue	Hazel	Green
Black	68	20	15	5
Brown	119	84	54	29
Red	26	17	14	14
Blond	7	94	10	16

```
[107]: df['Total'] = df.sum(axis=1)
df.loc['Total'] = df.sum(axis=0)
n = df.loc['Total', 'Total']
row_totals = df.loc['Total'].drop('Total')
col_totals = df['Total'].drop('Total')

expected = np.outer(col_totals, row_totals) / n

# Convert to DataFrame for clarity
expected_df = pd.DataFrame(expected, index=df.index[:-1], columns=df.columns[:-1])
display(expected_df.round(2))
```

	Brown	Blue	Hazel	Green
Black	40.14	39.22	16.97	11.68
Brown	106.28	103.87	44.93	30.92
Red	26.39	25.79	11.15	7.68
Blond	47.20	46.12	19.95	13.73

c)

```
[108]: cell_mass = (df_copy - expected_df)**2 / expected_df
display(cell_mass.round(2))
```

	Brown	Blue	Hazel	Green
Black	19.35	9.42	0.23	3.82
Brown	1.52	3.80	1.83	0.12
Red	0.01	2.99	0.73	5.21
Blond	34.23	49.70	4.96	0.38

```
[109]: chi2_stat = cell_mass.to_numpy().sum()
chi2_stat
```

```
[109]: 138.28984162600824
```

The degrees of freedom of the χ^2 is equal to $(\# \text{ of rows} - 1) \cdot (\# \text{ of columns} - 1) = 3 \cdot 3 = 9$.

$$\text{Total Intertia} = \frac{\chi^2}{9} \approx 15.37$$

d)

```
[89]: df_ca = df.drop(index='Total').drop(columns='Total')

ca = prince.CA(n_components=2, n_iter=10, copy=True, check_input=True,
    ↪random_state=42)
ca = ca.fit(df_ca)

row_coords = ca.row_coordinates(df_ca)
col_coords = ca.column_coordinates(df_ca)

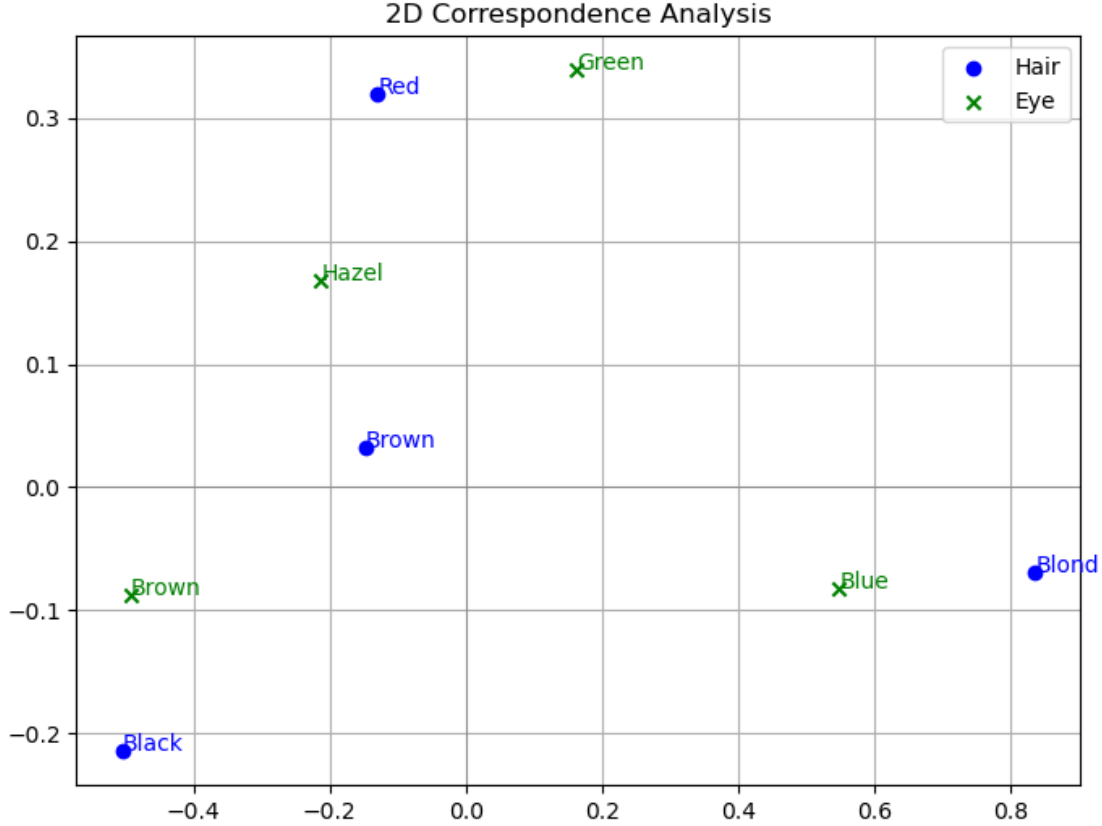
fig, ax = plt.subplots(figsize=(8, 6))
ax.scatter(row_coords[0], row_coords[1], c='blue', label='Hair')
for i, txt in enumerate(df_ca.index):
    ax.annotate(txt, (row_coords.iloc[i, 0], row_coords.iloc[i, 1]),
    ↪color='blue')

ax.scatter(col_coords[0], col_coords[1], c='green', marker='x', label='Eye')
for i, txt in enumerate(df_ca.columns):
    ax.annotate(txt, (col_coords.iloc[i, 0], col_coords.iloc[i, 1]),
    ↪color='green')

ax.axhline(0, color='grey', lw=0.5)
ax.axvline(0, color='grey', lw=0.5)
ax.set_title('2D Correspondence Analysis')
ax.legend()
plt.grid(True)
plt.show()

try:
    inertia = ca.explained_inertia_
except AttributeError:
    eigenvalues = ca.eigenvalues_
    total = sum(eigenvalues)
    inertia = [val / total for val in eigenvalues]

print("Explained inertia per dimension:", inertia)
print(f"Total captured in 2D: {sum(inertia):.2%}")
```



Explained inertia per dimension: [0.9037805836477717, 0.0962194163522283]

Total captured in 2D: 100.00%

The 2-dimensional correspondence analysis explains 100% of the total variation, with **Dimension 1 capturing 90.38% and Dimension 2 capturing 9.62%**. The biplot shows **strong associations between Blond hair and Blue eyes, as well as Red hair and Green eyes**. The category **Hazel eyes** appears more centrally positioned, indicating it is less strongly associated with any specific hair color. The proximity of **Black hair and Brown eyes also suggests a notable pairing**. These findings reveal meaningful deviations from independence between hair and eye color categories.

5. Conditional Multivariate Normal

a)

The general formula for conditional expectation of a multivariate normal random variable is given by:

$$\mathbb{E}[X_1|X_2 = x_2] = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

Using this, we can extend to include the second conditional $X_3 = x_3$:

$$\mathbb{E}[X_1|X_2 = x_2, X_3 = x_3] = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2) + \Sigma_{13}\Sigma_{33}^{-1}(x_3 - \mu_3)$$

b)

The general formula for conditional variance of a multivariate normal random variable is given by:

$$\text{Var}(X_1|X_2 = x_2) = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Using this, we can extend to include the second conditional $X_3 = x_3$:

$$\text{Var}(X_1|X_2 = x_2) = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} - \Sigma_{13}\Sigma_{33}^{-1}\Sigma_{31}$$

c)

We know that the sum of two multivariate normal distributions is also multivariate normal.

$$X_2 + X_3 \sim \mathcal{N}(\mu_2 + \mu_3, \Sigma_{22} + \Sigma_{33})$$

The reason that $\text{Var}(X_2 + X_3) = \Sigma_{22} + \Sigma_{33}$ is that $\Sigma_{23} = \Sigma_{32} = 0$, as given by the problem.

Now, the joint distribution of

$$\begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 + \mathbf{X}_3 \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu_1 \\ \mu_2 + \mu_3 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} + \Sigma_{13} \\ \Sigma_{21} + \Sigma_{31} & \Sigma_{22} + \Sigma_{33} \end{bmatrix} \right)$$

d)

We now derive the conditional distribution of $\mathbf{X}_1$ given that $\mathbf{X}_2 + \mathbf{X}_3 = \mathbf{x}_0$.

From the joint distribution,

$$\begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 + \mathbf{X}_3 \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu_1 \\ \mu_2 + \mu_3 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} + \Sigma_{13} \\ \Sigma_{21} + \Sigma_{31} & \Sigma_{22} + \Sigma_{33} \end{bmatrix} \right)$$

Using the formula for the conditional distribution of a multivariate normal, we get:

$$\mathbf{X}_1 | (\mathbf{X}_2 + \mathbf{X}_3 = \mathbf{x}_0) \sim \mathcal{N}(\mu_1 + (\Sigma_{12} + \Sigma_{13})(\Sigma_{22} + \Sigma_{33})^{-1}(\mathbf{x}_0 - \mu_2 - \mu_3), \Sigma_{11} - (\Sigma_{12} + \Sigma_{13})(\Sigma_{22} + \Sigma_{33})^{-1}(\Sigma_{21} + \Sigma_{31}))$$

[]: